

Experiment No. :05

Name of The Experiment: Study and Observation of TRIAC Characteristics

Objectives

- To display the characteristic curve of a TRIAC on the oscilloscope using X-Y mode.
- To investigate the relationship between the gate trigger voltage U_{GT} and the off-state forward voltage U_D .
- To observe the blocking and conduction behavior of the TRIAC in both positive and negative directions.
- To observe the TRIAC voltage and current waveforms in time-domain mode.
- To verify the bidirectional switching operation of the TRIAC.

Introduction

A TRIAC (Triode for Alternating Current) is a bidirectional thyristor used for controlling AC power. It can be considered as two thyristors connected in anti-parallel with their gates joined together. Therefore, unlike a conventional thyristor, a TRIAC can conduct current in both directions.

The two main terminals of a TRIAC are called MT1 and MT2, while the third terminal is the gate (G). The TRIAC remains in the off-state or blocking state until it is triggered. Triggering can occur when an appropriate gate current is applied or when the voltage across the main terminals reaches the breakover voltage.

Once triggered, the TRIAC changes from the blocking state to the conducting state. It remains conducting as long as the current through the device remains above its holding current. Increasing the gate trigger voltage U_{GT} generally allows the TRIAC to turn on at a lower off-state voltage U_D .

The characteristic curve of a TRIAC is approximately symmetrical about the origin. It consists of a blocking region followed by a rapid transition into the conducting state in both positive and negative directions. Thus, TRIAC is suitable for AC switching and power-control applications such as lamp dimming and AC voltage control.

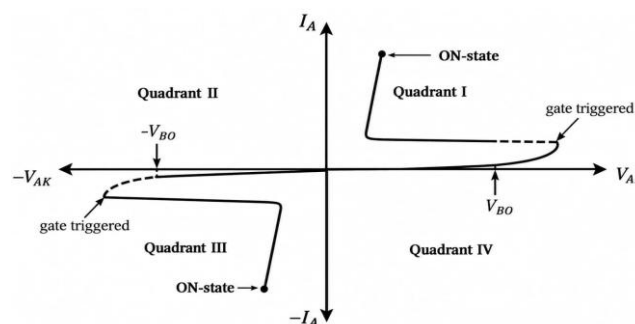


Figure 1: Characteristic curve of a TRIAC

Apparatus

Main Equipment

- Power Supply PTE-047-01
- Characteristic PTE-047-03 (TRIAC BT137 module, 12 V / 2 W lamp, $R_1 = 100\ \Omega$, $R_2 = 100\ \text{k}\Omega$ potentiometer, $R_3 = 10\ \Omega$)
- Container Box Section PTE-047-08

Auxiliary Equipment

- Digital Multimeter
- Digital Storage Oscilloscope (Protek 5100, 100 MHz, 500 MSa/s)

Circuit Diagram

The experimental circuit consists of a 12 V AC supply, a BT137 TRIAC, a 12 V / 2 W lamp, and the associated triggering and sensing components. The TRIAC is connected in series with the lamp as the AC load.

The gate of the TRIAC is triggered through the $100\ \text{k}\Omega$ potentiometers R_2 and the $100\ \Omega$ current-limiting resistors R_1 . The gate trigger voltage U_{GT} is measured using a digital multimeter.

The voltage across MT2 and MT1 of the TRIAC is connected to Channel 1 (CH1) of the oscilloscope and represents the X-axis in X-Y mode. The voltage across the $10\ \Omega$ sense resistor R_3 is connected to Channel 2 (CH2) and represents the Y-axis. Since the value of R_3 is known, the TRIAC current can be calculated using Ohm's law:

$$I_{TRIAC} = \frac{U_{R3}}{R_3}$$

Thus, the oscilloscope can be used to obtain the voltage-current characteristic of the TRIAC.

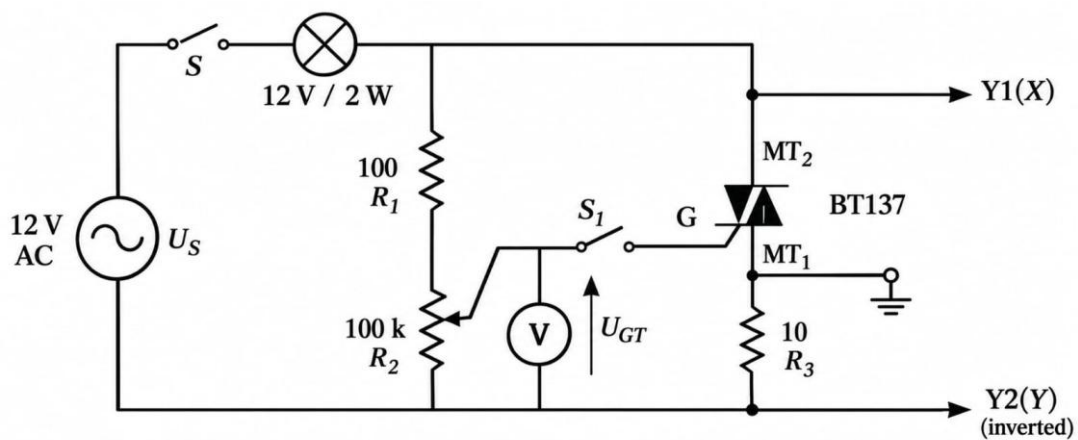


Figure 2: Circuit diagram for obtaining the TRIAC characteristic

Procedure

1. Connect the Power Supply, TRIAC Characteristic Module, and Container Box panels according to the experimental circuit diagram.
2. Set the oscilloscope to X-Y mode.
3. Connect Channel 1 (CH1) across the TRIAC between MT2 and MT1 to represent the X-axis.
4. Connect Channel 2 (CH2) across the $10\ \Omega$ sense resistor R_3 to represent the Y-axis. Set the channel to inverted mode as required.
5. Apply the 12 V AC supply to the circuit and close the load switch so that the lamp is connected.
6. Close switch S_1 and trigger the TRIAC by adjusting the potentiometer R_2 .
7. Record the gate trigger voltage U_{GT} , the off-state voltage U_D , and the voltage across the sense resistor U_{R3} .
8. Repeat the measurements for different settings of the potentiometer and record the corresponding values.
9. Capture the TRIAC characteristic curve in X-Y mode for the different trigger conditions.
10. Switch the oscilloscope to normal time-domain mode and observe the voltage across the TRIAC and the voltage across R_3 .
11. Record the peak-to-peak voltage and period of the observed waveforms.
12. Calculate the TRIAC current using:

$$I_{TRIAC} = \frac{U_{R3}}{R_3}$$

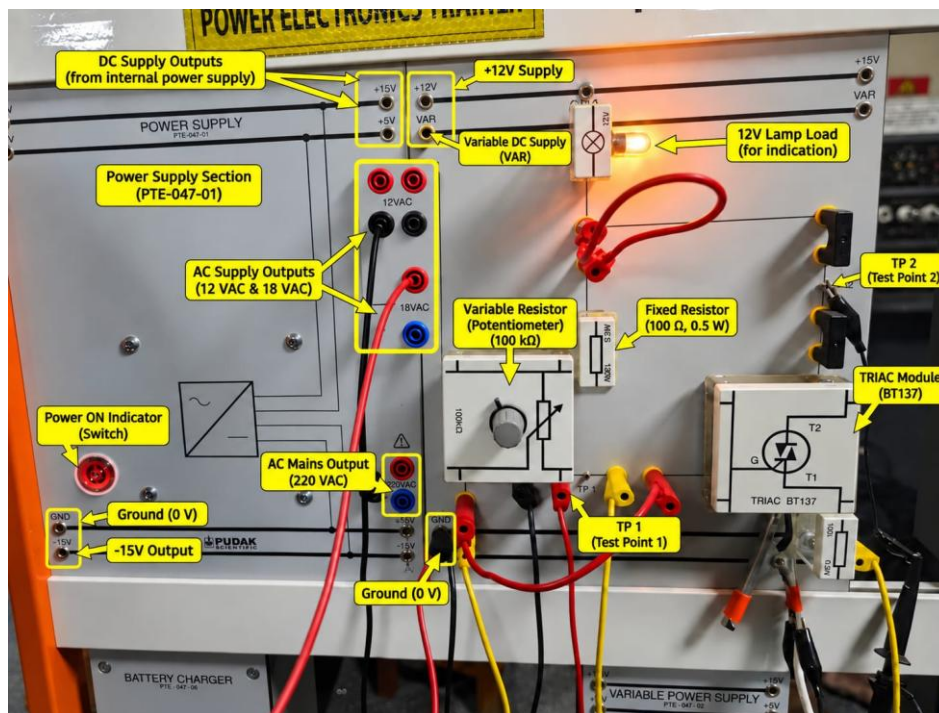


Figure 3: Circuit Diagram

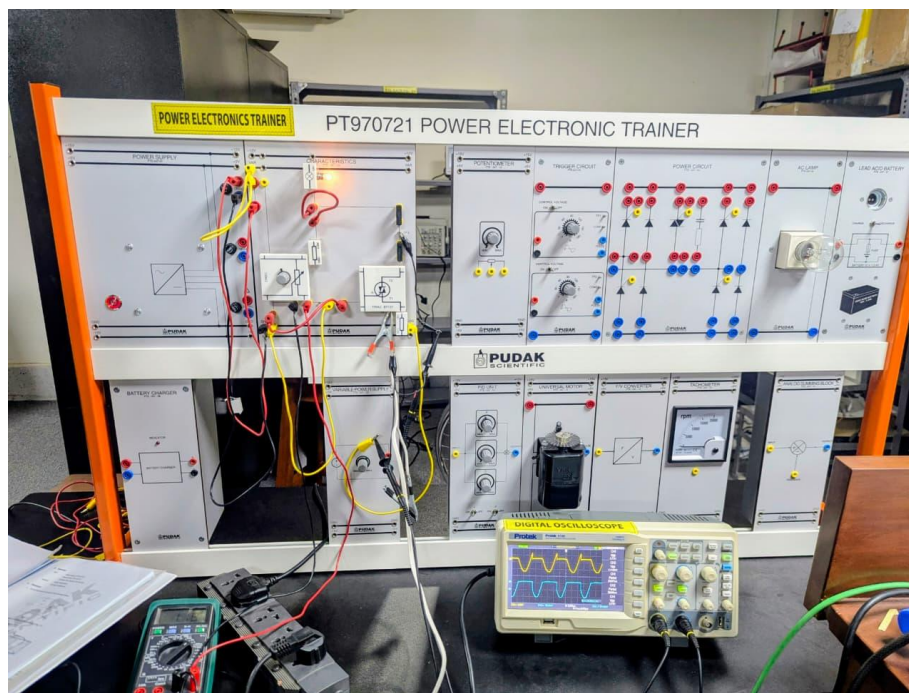


Figure 4: Complete experimental setup with the Puduk PT970721 trainer, multimeter and digital oscilloscope

Data Table

No.	$U_{GT}(V)$	$U_D(V)$	$U_{R3}(V)$	$I_{TRIAC}(A)$
1	1.70	3.52	3.08	0.308
2	1.24	2.44	1.84	0.184
3	1.77	1.02	3.16	0.316
4	1.60	2.30	3.32	0.332
5	1.76	1.18	3.12	0.312

Table 1: Experimental data

Oscillograms in X-Y Mode

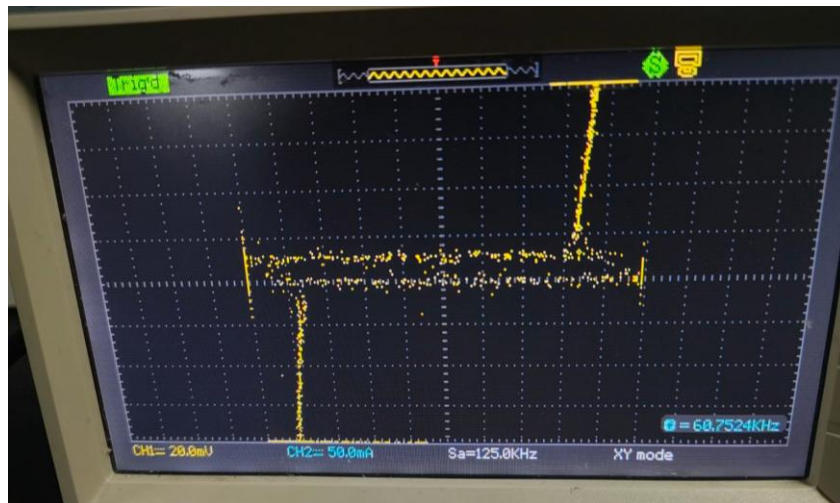


Figure 5: TRIAC characteristic in X-Y mode (CH1 = 20 mV/div, CH2 = 50 mA/div)



Figure 6: TRIAC characteristic in X-Y mode (CH1 = 20 mV/div, CH2 = 200 mA/div)

Oscillograms in Time-Domain Mode

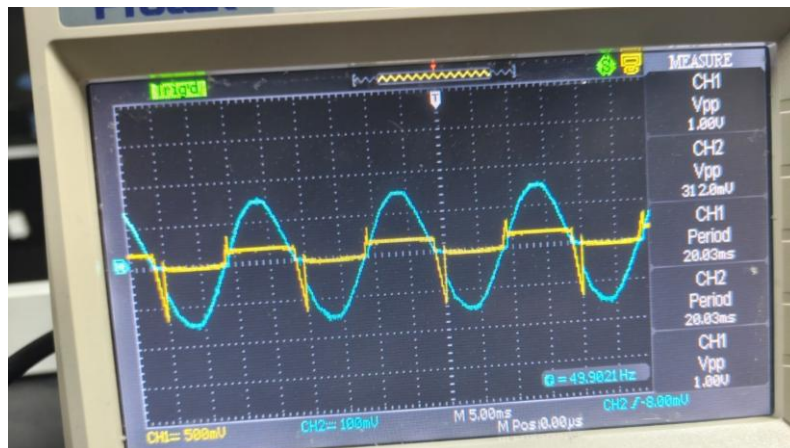


Figure 7: Voltage across TRIAC (CH1) and across R3 (CH2), CH1 $V_{pp} = 1.00$ V



Figure 8: Voltage across TRIAC (CH1) and across R3 (CH2), CH1 $V_{pp} = 1.18$ V

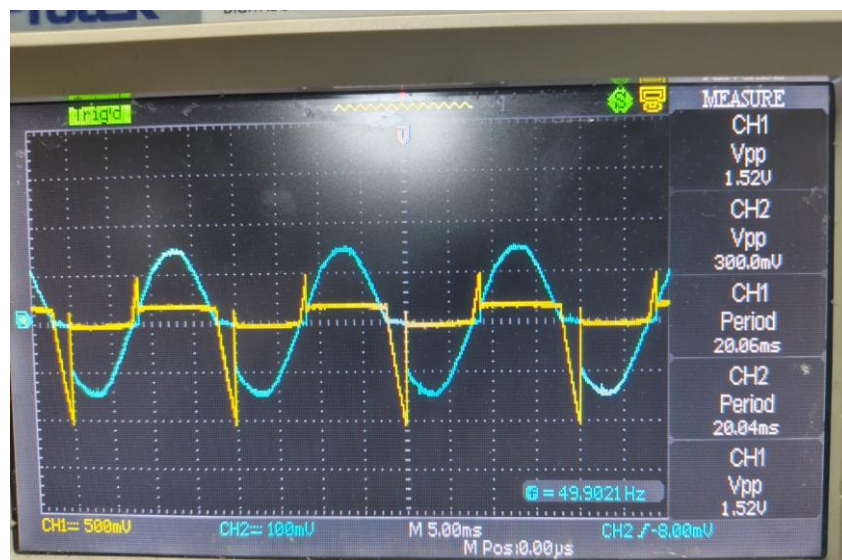


Figure 9: Voltage across TRIAC (CH1) and across R3 (CH2), CH1 $V_{pp} = 1.52$ V

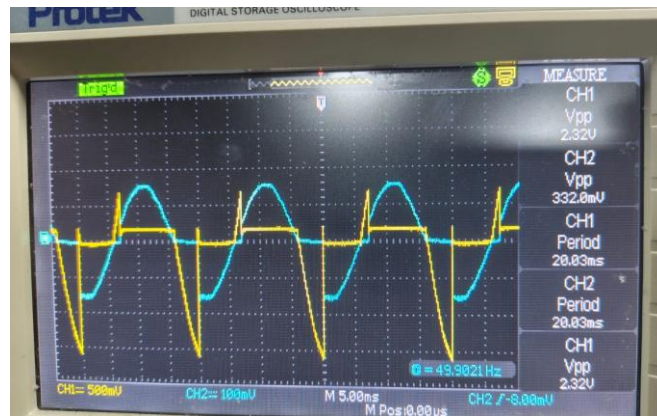


Figure 10: Voltage across TRIAC (CH1) and across R3 (CH2), CH1 $V_{pp} = 2.32 \text{ V}$

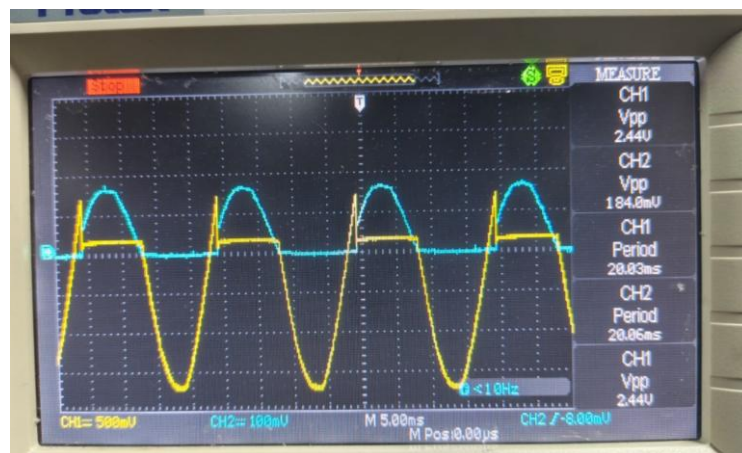


Figure 11: Voltage across TRIAC (CH1) and across R3 (CH2), CH1 $V_{pp} = 2.44 \text{ V}$ (largest U_D)

Table 2: Peak-to-peak values read from the oscilloscope measurements

Figure	CH1 $V_{pp}(\text{V})$	CH2 $V_{pp}(\text{mV})$	Period (ms)	$I_{pp} = V_{pp}/R_3(\text{mA})$
Figure 7	1.00	312	20.03	31.2
Figure 8	1.18	320	20.05	32.0
Figure 9	1.52	300	20.06	30.0
Figure 10	2.32	332	20.03	33.2
Figure 11	2.44	184	20.03	18.4

Discussion

From Table 1, when U_{GT} is low (1.24 V), the TRIAC needs a large off-state voltage before it fires ($U_{AK} = 2.44$ V). When U_{GT} is increased to 1.60 V, 1.76 V and 1.77 V, the off-state voltage at which the TRIAC fires decreases to 2.30 V, 1.18 V and 1.02 V respectively. This agrees with the theory that a stronger gate drive lets the TRIAC switch on at a lower forward voltage, so the blocking region of the characteristic becomes shorter.

The reading $U_{GT} = 1.70$ V with $U_{AK} = 3.52$ V does not follow this trend. It is also higher than the largest CH1 value captured on the oscilloscope (2.44 V), and one capture reads 1.52 V (Figure 9). This point is therefore likely to be a reading or recording error, and it should be rechecked if the experiment is repeated.

The X-Y oscillograms (Figures 5 and 6) confirm the symmetrical behaviour of the TRIAC. The device blocks in the off-state with almost zero current, then switches to a low-voltage on-state in both the positive (first quadrant) and negative (third quadrant) directions. This differs from a thyristor, which conducts in only one direction. The time-domain captures also show that the TRIAC conducts in both half-cycles of the 50 Hz supply, with the flat on-state voltage visible once it has been triggered.

Possible sources of error include noise and trace dispersion in X-Y mode (visible as scattered points in Figures 5 and 6), contact resistance in the patch leads, limited resolution of the potentiometer setting, and reading errors when taking values from the oscilloscope display.

Conclusion

The characteristic behavior of the BT137 TRIAC was successfully observed using an oscilloscope in both X-Y and time-domain modes. The X-Y characteristic demonstrated the blocking and conduction regions of the TRIAC in both positive and negative directions, confirming its bidirectional switching behavior.

The experimental observations also showed that an increase in gate trigger voltage U_{GT} generally reduces the off-state voltage U_D required to trigger the TRIAC. The time-domain waveforms further confirmed the transition of the TRIAC from the blocking state to the conducting state during the AC cycle.

Therefore, the experiment successfully demonstrated the fundamental voltage-current characteristics and switching operation of a TRIAC in an AC circuit.

References

- [1] Rashid, Muhammad H., Power Electronics: Circuits, Devices, and Applications, Prentice-Hall International, Inc., New Jersey, 1998.
- [2] Pudak Scientific, Characteristics of TRIAC (LE30003E), Power Electronics Trainer PT970721 Laboratory Manual.